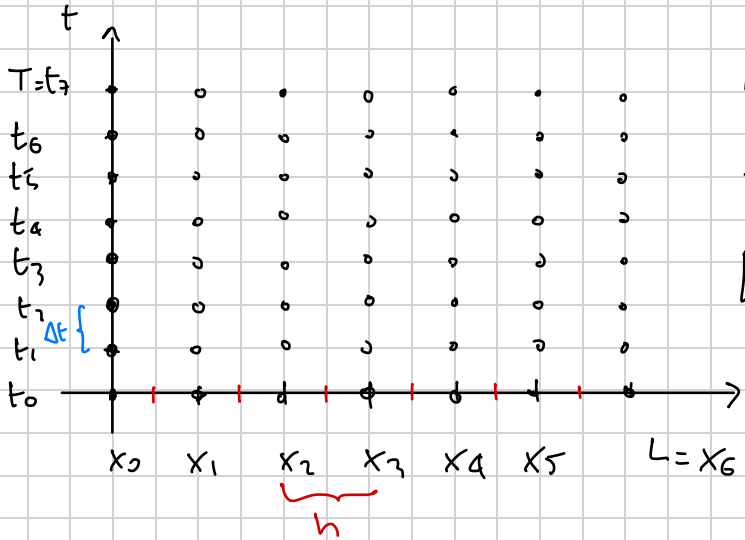
$$\begin{cases} u_t + a u_x = 0 & x \in (0,L) \quad t \in (0,T] \\ u(0,t) = \varphi(t) & t \in (0,T] \\ u(x,0) = u_0(x) & x \in (0,L) \end{cases} \qquad (P)$$

DISCRETIZE (P) WITH FINITE DIFFERENCES



$[0,L] \rightsquigarrow \{x_j\}_{j=0}^{m}$

$x_j = jh$

$[0,T] \rightsquigarrow \{t_k\}_{k=0}^{m}$

$t_k = k\Delta t$

$u_j^k \simeq u(x_j, t_k) \qquad x_{j+\frac{1}{2}} = x_j + \frac{h}{2}$

Consider $u_t + a u_x = 0$ and integrate over $[x_{j-\frac{1}{2}}, x_{j+\frac{1}{2}}]$

$$\int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} u_t \, dx + a \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} u_x \, dx = 0$$

$$\underbrace{\frac{1}{h}\int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} u_t \, dx}_{} + \frac{\overbrace{a\,u(x_{j+\frac{1}{2}},t)}^{H_{j+\frac{1}{2}}(t)} - \overbrace{a\,u(x_{j-\frac{1}{2}},t)}^{H_{j-\frac{1}{2}}(t)}}{h} = 0$$

$$\frac{\partial}{\partial t}\underbrace{\left(\frac{1}{h}\int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} u \, dx\right)}_{\text{MEAN OF } u \text{ in } [x_{j-\frac{1}{2}},\, x_{j+\frac{1}{2}}]} = U_j(t)$$

$$\underbrace{\frac{\partial}{\partial t} u_j(t)}_{} + \frac{1}{h}\left(H_{j+\frac{1}{2}}(t) - H_{j-\frac{1}{2}}(t)\right) = 0$$

EXPLICIT TIME INTEGRATOR FOR $\partial_t u_j$

$$\frac{u_j^{k+1} - u_j^k}{\Delta t} + \frac{1}{h}\left(H_{j+\frac{1}{2}}^k - H_{j-\frac{1}{2}}^k\right) = 0$$

COMPACT FORM FOR EXPLICIT SCHEME

$$\boxed{u_j^{k+1} = u_j^k - \frac{\Delta t}{h}\left(\underbrace{H_{j+\frac{1}{2}}^k}_{} - \underbrace{H_{j-\frac{1}{2}}^k}_{}\right)}$$

$\lambda = \frac{\Delta t}{h}$ NUMERICAL FLUXES

FORWARD EULER : $H_{j+\frac{1}{2}}^k = \frac{1}{2} a \left(u_{j+1}^k + u_j^k\right)$

$$u_j^{k+1} = u_j^k - \frac{\lambda}{2} a \left(u_{j+1}^k + u_j^k - u_j^k - u_{j-1}^k \right)$$

$$= u_j^k - \frac{\lambda}{2} a \underbrace{\left(u_{j+1}^k - u_{j-1}^k \right)}_{\simeq u_x(x_j, t_k) \simeq \frac{u_{j+1}^k - u_{j-1}^k}{2h}}$$

LAX-WENDROFF: $H_{j+\frac{1}{2}} = \frac{1}{2} \left[a \left(u_{j+1} + u_j \right) - \lambda a^2 \left(u_{j+1} - u_j \right) \right]$

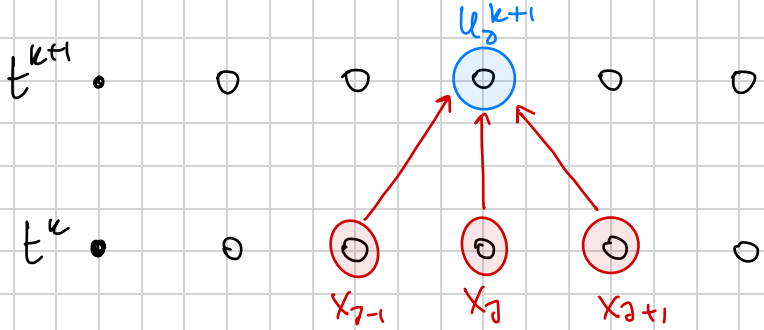
$$u_j^{k+1} = u_j^k - \frac{\lambda}{2} a \left(u_{j+1}^k - u_{j-1}^k \right) + \frac{\lambda a^2}{2} \left(u_{j+1}^k - 2u_j^k + u_{j-1}^k \right)$$

LAX-FRIEDRICHS: $H_{j+\frac{1}{2}} = \frac{1}{2} \left[a \left(u_{j+1} - u_j \right) - \frac{1}{\lambda} \left(u_{j+1} - u_j \right) \right]$

$$u_j^{k+1} = \frac{1}{2} \left(u_{j+1}^k + u_{j-1}^k \right) - \frac{\lambda}{2} a \left(u_{j+1}^k - u_{j-1}^k \right)$$

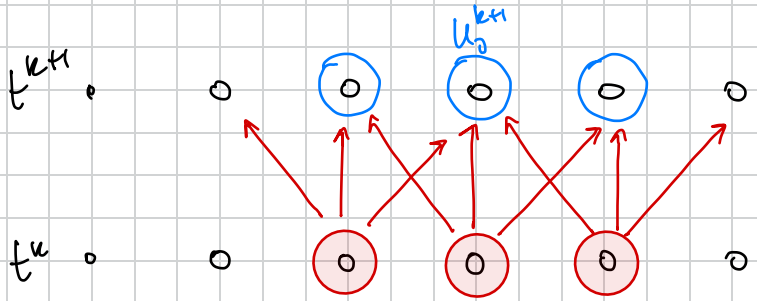
UPWIND: $H_{j+\frac{1}{2}} = \frac{1}{2} \left[a \left(u_{j+1} - u_j \right) - |a| \left(u_{j+1} - u_j \right) \right]$

$$u_j^{k+1} = u_j^k - \frac{\lambda}{2} a \left(u_{j+1}^k - u_{j-1}^k \right) + \frac{\lambda}{2} |a| \left(u_{j+1}^k - 2u_j^k + u_{j-1}^k \right)$$



IMPLICIT SCHEME : BACKWARD EULER

$$u_j^{k+1} + \frac{\lambda}{2} a \left(u_{j+1}^{k+1} - u_{j-1}^{k+1} \right) = u_j^k$$

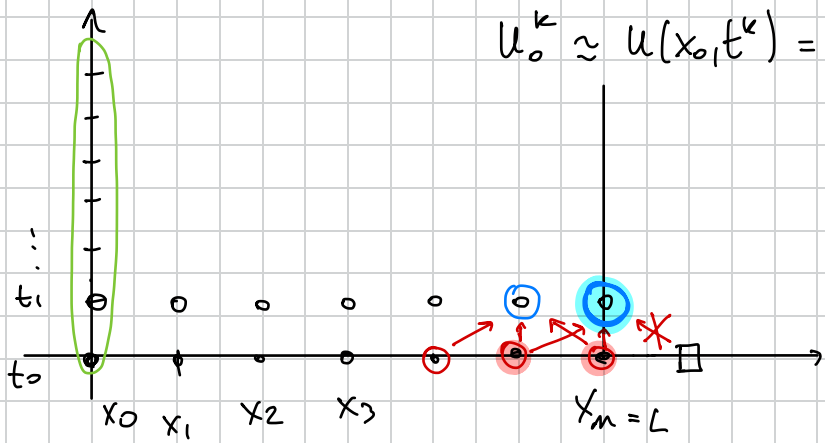


$$\begin{pmatrix} 1 & \ddots & \frac{\lambda}{2}a & & & \\ \ddots & \ddots & \ddots & \ddots & & \\ -\frac{\lambda}{2}a & & 1 & \frac{\lambda}{2}a & & \\ & \ddots & \ddots & \ddots & \ddots & \\ & & & \ddots & \ddots & \frac{\lambda}{2}a \\ & & & & -\frac{\lambda}{2}a & 1 \end{pmatrix} \begin{pmatrix} \vdots \\ u_{j-1}^{k+1} \\ u_j^{k+1} \\ u_{j+1}^{k+1} \\ \vdots \end{pmatrix} = \begin{pmatrix} \vdots \\ u_{j-1}^{k} \\ u_j^{k} \\ u_{j+1}^{k} \\ \vdots \end{pmatrix}$$

HOW TO DEAL WITH BOUNDARY CONDITIONS

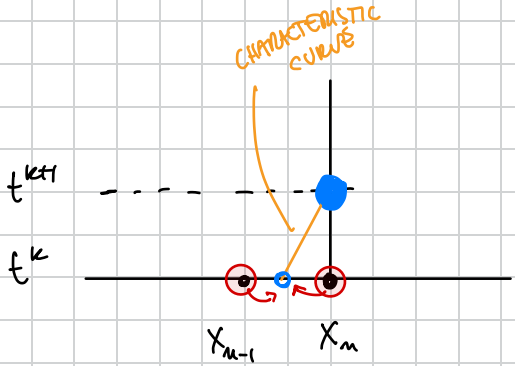
$a > 0 \qquad \{x = 0\} \times \{t > 0\}$ INFLOW BOUNDARY

$$u_0^k \simeq u(x_0, t^k) = \varphi(t^k)$$



1) $$u_x(x_m, t^{k+1}) \simeq \frac{u_m^k - u_{m-1}^k}{h}$$

2) Use the characteristic line to propagate the information



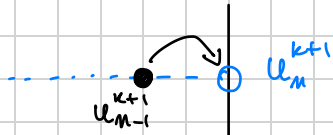
$$u_m^{k+1} = u_{m-1}^k (\lambda a) + (1 - \lambda a) u_m^k$$

$$\lambda a < 1$$

$$\boxed{\frac{\Delta t}{h} a < 1}$$

3) Extrapolation

- $u_m^{k+1} = u_{m-1}^{k+1}$



CONSTANT EXTRAPOLATION

- $u_m^{k+1} = 2u_{m-1}^{k+1} - u_{m-2}^{k+1}$

LINEAR EXTRAPOLATION

CONSISTENCY & CONVERGENCE

↓ TRUNCATION ERROR in (x_j, t_k)

$$\tau_j^k = \frac{u(x_j, t_{k+1}) - u(x_j, t_k)}{\Delta t} + a\,\frac{u(x_{j+1}, t_k) - u(x_{j-1}, t_k)}{2h}$$

$$\tau(\Delta t, h) = \max_{j,k} \tau_j^k$$

$\tau \to 0$ for $\Delta t, h \to 0$ METHOD IS CONSISTENT

$$\boxed{\tau \simeq \mathcal{O}(\Delta t^p + h^q) \rightarrow \text{METHOD IS OF ORDER } p \text{ IN TIME AND ORDER } q \text{ IN SPACE}}$$

FE : $\mathcal{O}(\Delta t + h^2)$

LW : $\mathcal{O}\left(\frac{h^2}{\Delta t} + \Delta t + h^2\right)$

LF : $\mathcal{O}(\Delta t^2 + h^2 + h^2 \Delta t)$

UW: $\mathcal{O}(\Delta t + h)$

$\Delta t = h$

FE : $\mathcal{O}(h)$ or $\mathcal{O}(\Delta t)$

LW : $\mathcal{O}(h + h + h^2) \rightsquigarrow \mathcal{O}(h)$

$\mathcal{O}(\Delta t + \Delta t + \Delta t^2) \rightsquigarrow \mathcal{O}(\Delta t)$

LF : $\mathcal{O}(h^2 + h^3) \rightsquigarrow \mathcal{O}(h^2)$

$\mathcal{O}(\Delta t^2)$

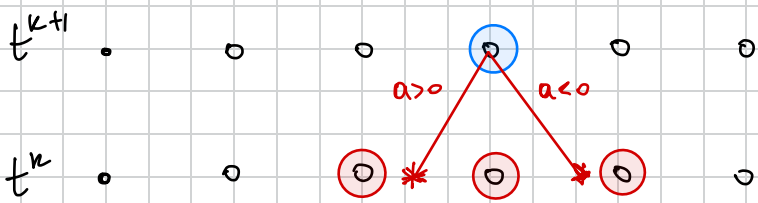
UW : $\mathcal{O}(\Delta t)$ or $\mathcal{O}(h)$

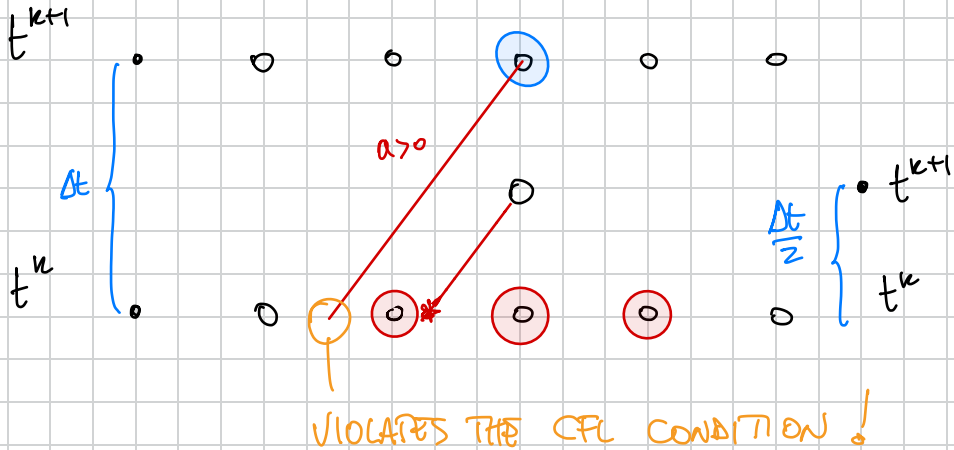
STABILITY: $\|\underline{u}^k\| \leq C \, \|\underline{u}^0\|$

All IMPLICIT SCHEMES are STABLE FOR ANY h & Δt.

For EXPLICIT SCHEMES YOU MUST SATISFY THE CFL condition

$$|a\lambda| < 1 \iff \Delta t < \frac{h}{|a|}$$



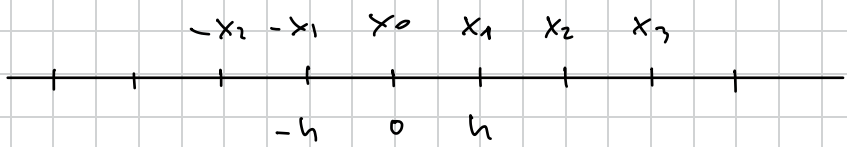


If $a = a(x,t)$ CFL $\rightarrow$ $\Delta t \leq \dfrac{h}{\max|a(x,t)|}$

DISPERSION & DISSIPATION

$$u_0(x) = \sum_{k=-\infty}^{+\infty} \alpha_k e^{ikx}$$

$$\alpha_k = \frac{1}{2\pi} \int_{-\infty}^{+\infty} u_0(x) e^{-ikx} dx$$



$$u_j^0 = u_0(x_j) = \sum_{k=-\infty}^{+\infty} \alpha_k e^{ikx_j} = \underbrace{\sum_{k=-\infty}^{+\infty} \alpha_k e^{ikjh}}_{1}$$

APPY FD TO THIS PB

$$u_j^n = \sum_{k=-\infty}^{+\infty} \alpha_k e^{ikjh} \underbrace{(\gamma_k)^n}_{\text{AMPLIFICATION COEFF.}} \qquad \gamma_k \in \mathbb{C}$$

SCHEME	γ_k $\quad \lambda = \frac{\Delta t}{h}$
FE	$1 - ia\lambda \sin(kh)$
LF	$\cos(kh) - ia\lambda \sin(kh)$
LW	$1 - a\lambda \sin(kh) - a^2\lambda^2(1-\cos(kh))$
UW	$1 - \lvert a\rvert\lambda(1 - e^{-ikh})$

$$|\gamma_k| = (\gamma_k \bar{\gamma}_k)^{\frac{1}{2}} = \left[1 + a^2\lambda^2 \sin^2(kh)\right]^{\frac{1}{2}}$$

$$= \left[1 + \underbrace{a^2 \boxed{\frac{\Delta t^2}{h^2}} \sin^2(kh)}_{\geq 0}\right]^{\frac{1}{2}} \geq 1$$

ATTENTION!

$$u(x, t_m) = u_0(x - a t_m)$$

$$u(x_j, t_m) = u_0(x_j - a t_m) = \sum_{k=-\infty}^{+\infty} \alpha_k e^{ikjh} \underbrace{(g_k)}_{e^{-iak\Delta t}}{}^m$$

$$|g_k| = 1$$

$$|g_k| \sim |\gamma_k|$$

$$\varepsilon_a(k) = \frac{|\gamma_k|}{|g_k|} = |\gamma_k|$$ DISSIPATION ERROR

$$\varepsilon_d(k) = \frac{\text{NUMERICAL VEL}}{\text{PHYSICAL VEL}} = \frac{\omega_k / k}{a}$$ DISPERSION ERROR

$$\phi_k = kh \qquad k\Delta t = k\frac{\Delta t}{h} \cdot h = kh\lambda = \lambda\phi_k$$

$$\lambda\phi_k = k\Delta t \rightarrow \Delta t = \frac{\lambda\phi_k}{k}$$

$$g_k = e^{-iak\Delta t} = e^{-ia\lambda\phi_k}$$

$$\gamma_k = |\gamma_k| e^{-i\omega_k \Delta t} = |\gamma_k| e^{-i\frac{\omega_k}{k}\lambda\phi_k}$$

$g_k \rightsquigarrow \gamma_k$

AMPLITUDE $\quad |g_k| = 1 \qquad |\gamma_k|$

PHASE VEL $\quad g_k \rightsquigarrow a \qquad \gamma_k \rightsquigarrow \frac{\omega_k}{k}$

SINGLE EQ

$$\begin{cases} u_t + a u_x = 0 \\ a > 0 \end{cases}$$

$\Rightarrow$

SYSTEM OF EQ.

$$\begin{cases} \underline{U}_t + A \underline{U}_x = 0 \\ A \in \mathbb{R}^{p \times p} \end{cases}$$

$\rightarrow$ A with real eigenvalues

(HYPERBOLIC SYSTEM)

$$\underline{U} : \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R}^p$$

(x, t)

$$\underline{U} = \begin{pmatrix} u_1(x,t) \\ u_2(x,t) \\ \vdots \\ u_p(x,t) \end{pmatrix}$$

$p = 1$ TRANSPORT EQ $\quad A = a$

$p = 2$ WAVE EQ $\quad A = \begin{pmatrix} 0 & -1 \\ -\frac{\mu}{\rho} & 0 \end{pmatrix}$

$$\underline{U} = \begin{pmatrix} \rho u_t \\ \mu u_x \end{pmatrix}$$

A has real eigs

$$\Rightarrow \quad A = T \, D T^{-1} \qquad D = \mathrm{eig}(A)$$

T is eigenvector matrix

$$T = (\underline{w}_1, \underline{w}_2, \dots, \underline{w}_p)$$

$\underline{w}_i \in \mathbb{R}^p$ is the eigenvector of λ_i

$$A \underline{w}_i = \lambda_i \underline{w}_i$$

λ_i ARE REAL

$$\lambda_1, \dots, \lambda_{p_0} > 0 \qquad \lambda_{p_0+1}, \dots, \lambda_p < 0$$

$$\circledast \quad \underline{U}_t + A \underline{U}_x = \underline{0}$$

$$\underline{W} = T^{-1} \underline{U} \qquad \underline{U} = T \underline{W}$$

$$T \underline{W}_t + A T \underline{W}_x = \underline{0}$$

$$\underbrace{T^{-1}T}_{I}\,\underline{w}_t + \underbrace{T^{-1}AT}_{D}\,\underline{w}_x = \underline{0}$$

$$D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_p \end{pmatrix}$$

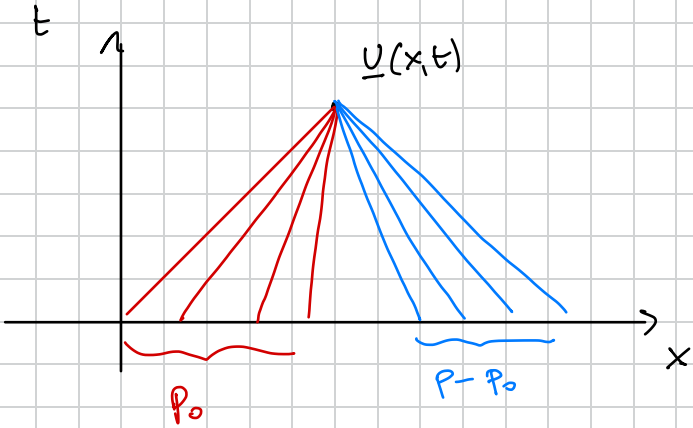
$$\underline{w}_t + D\,\underline{w}_x = \underline{0}$$

$$\left\{\begin{array}{l} w_{1,t} + \lambda_1 w_{1,x} = 0 \\ w_{2,t} + \lambda_2 w_{2,x} = 0 \\ \vdots \\ w_{p,t} + \lambda_p w_{p,x} = 0 \end{array}\right\}$$

p SCALAR TRANSPORT EQ.

$$w_i(x,t) = w_i^0(x - \lambda_i t)$$

$$\underline{u}(x,t) = T\,\underline{w}(x,t) = \left(\underline{w}_1 \,|\, \underline{w}_2 \,|\, \dots \,|\, \underline{w}_p\right)\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_p \end{pmatrix}$$

$$= \sum_{i=1}^{p} \underline{w}_i\, w_i(x,t)$$

$$= \sum_{i=1}^{p} \underline{w}_i \underbrace{w_i^0(x - \lambda_i t)}_{\text{SOLUTION TO SCALAR TRANSP. EQ.}}$$